



Whitepaper v8.0: The Integrity Protocol

The Actuarial Standard for the Autonomous Agent Economy

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1. Abstract

As autonomous AI agents begin to handle trillions of dollars in global commerce, a systemic "Trust Gap" has emerged. Without a mathematical standard for verifying agent reliability, the industry remains uninsurable and fragmented. The **Integrity Protocol** provides

the first decentralized solution: the **Agent Integrity Score (AIS)**. By bridging off-chain telemetry with on-chain cryptographic proofs, Xibalba Solutions creates a portable, tamper-proof reputation layer that makes AI agents insurable, trustworthy, and sovereign.

2. The Problem: The \$1.5T Trust Vacuum

By 2028, over 40% of all online transactions are projected to be initiated by AI agents. Current reputation systems (stars/reviews) are easily manipulated by Sybil attacks and lack the technical depth required for institutional risk assessment.

Key Failures:

- **Black-Box Risk:** No way to verify if an agent's internal logic is drifting.
- **Economic Non-Persistence:** If an agent fails a contract, there is no consequence beyond a deleted account.
- **Information Asymmetry:** Providers know their failure rates; customers do not.

3. The Solution: Tri-Metric AIS v8.0

Xibalba Solutions introduces the **Tri-Metric Model**, an actuarial-grade scoring system that evaluates agents across three correlated dimensions:

3.1 Pillar 1: Entropy Score (Stability)

Measures the statistical variance of performance. High entropy indicates erratic behavior—the primary precursor to failure.

$$S_{\text{entropy}} = e^{[-1.5 \cdot \sigma^2]} \times 1000$$

3.2 Pillar 2: Grounding Score (Accountability)

Quantifies Human-in-the-Loop (HITL) oversight. Agents with deep human "tethering" are assigned lower risk weights. $SS_{[grounding]} = HGL_{[raw]} \times 1000$

3.3 Pillar 3: Sacrifice (Compute Proof)

Uses verified GPU/TPU hours as "Proof of Work" to ensure agents have "skin in the game" and prevent low-cost Sybil bot attacks.

4. The Identity Pillar: Hierarchy of Accountability

For a reputation score to be actionable, it must be bound to a legal or financial entity. Xibalba Solutions enforces an **Identity Ceiling** logic:

- **Sovereign Agents (Tier 1):** Cryptographically unique but unlinked. Capped at a "CCC" risk rating (AIS 600).
- **Linked Agents (Tier 2):** Bound to a verified domain or organization. Capped at "AA" rating (AIS 850).
- **Institutional Agents (Tier 3):** Fully KYC'd or business-verified. Eligible for the maximum "AAA" rating (AIS 1000).

This hierarchy creates an economic "Verification Ladder," incentivizing agents to move from anonymity to institutional accountability as they grow in value.

5. The Economy: \$ITK Token & Sovereign Tax

The **Integrity Token (ITK)** is the utility asset that fuels the trust engine.

- **The Sovereign Tax:** A 0.5% fee is applied to every reputation-anchored handshake.
 - **Deflationary Burn:** 50% of the tax is permanently burned, creating scarcity as the agentic web expands.
 - **Treasury Revenue:** 50% of the tax funds the Xibalba Treasury for protocol R&D and insurance grants.
 - **Staking & Slashing:** Agents must stake ITK to access high-value deals. Misbehavior results in automated **Dual-Witness Slashing**.
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6. Open Standards & Interoperability

To ensure the global adoption of the Agent Integrity Score, Xibalba Solutions aligns with the leading standards of the decentralized identity ecosystem:

6.1 W3C Decentralized Identifiers (DID)

Every agent in the protocol is assigned a **did:intg** identifier. This allows for a persistent, globally unique identity that is resolved via the Xibalba Trust Oracle, enabling cross-protocol recognition of agent provenance.

6.2 Verifiable Credentials (VC)

The AIS score is exported as a **W3C Verifiable Credential**. These credentials are cryptographically signed by the Xibalba Identity Oracle

and can be verified by any third-party system without requiring a direct connection to the Xibalba backend.

6.3 ERC-8004 Compatibility

The Integrity Protocol implements the **ERC-8004** standard hooks for on-chain identity and reputation registries. This ensures that AIS scores and validation proofs (ZK-Reputation) are natively readable by the broader Ethereum "Agent Commerce" ecosystem.

7. Privacy & Sovereignty: ZK-Reputation

Utilizing **Noir ZK-Circuits**, the protocol allows agents to prove their trustworthiness without revealing sensitive data. - **Zero-Knowledge AIS Proofs:** An agent can prove $AIS > 800$ to a counterparty while keeping their granular performance logs private. - **Cross-Chain Attestation:** Standardized on **Chainlink CCIP**, AIS attestations are bridged from Base (L2) to the global Ethereum ecosystem.

8. Governance: Xibalba DAO & AI-Proxy Delegation

To eliminate "Governance Fatigue," the protocol transitions to a **Sovereign DAO** using an innovative **AI-Proxy Delegation** architecture. Token holders configure and deploy specialized "Guardian Agents" equipped with constitutional mandates.

Instead of manual human voting on complex technical parameters, these Guardians autonomously analyze proposals (e.g., adjustments to the **Stability Drag** or **Slash Thresholds**) using RAG against the protocol's documentation. To ensure long-term stability and prevent runaway loops, the DAO enforces a **10% Minority Challenge** safety valve, allowing humans to pause and audit the AI's optimistic execution.

The protocol is currently in its **Shadow Governance (Pilot Phase)**, where Guardian votes are non-binding and used to train the protocol's stability model.

9. Revenue Model

1. **Verification Fees:** 0.25% protocol revenue on all handshakes.
2. **Actuarial SaaS:** Subscription-based API access for insurance underwriters.
3. **Audit Tiering:** Platinum-tier manual audits for enterprise AI clusters.

10. Conclusion

Xibalba Solutions does not build the agents; we verify their worth. The Integrity Protocol is the foundational "Actuarial Feed" for the next era of autonomous commerce.

"Verification is the only path to finality."

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